INCH-POUND

MIL-S-13949/48
16 August 1993
SUPERSEDING
MIL-P-13949/4C
11 February 1987

MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, LAMINATED, BASE MATERIAL GF (WOVEN E-GLASS REINFORCEMENT, MAJORITY BIFUNCTIONAL EPOXY RESIN, FLAME RESISTANT, METAL CLAD OR UNCLAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: G in accordance with MIL-S-13949.

Resin system: F in accordance with MIL-S-13949.

Special consideration (third letter designator): G, K, N, P or T in accordance with MIL-S-13949.

Type of metal foil: Any as specified in accordance with MIL-S-13949.

Weight of metal foil: E, Q, T, H, M, 1, 2, 3, 4, 5 or 0 in accordance with MIL-S-13949.

Solderability: Base materials clad with metal foil types A, B, C, D, G, N, J, K, L, and M shall be tested in accordance with MIL-S-13949. Solderability testing not applicable on all other metal foil types, unless otherwise specified in the ordering data.

Bow and twist: C or X in accordance with MIL-S-13949.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values:

Dimensional stability (for base material thicknesses less than .020 inch (0.51 mm)).

After bake: Class A, .00075 inch/inch (0.00075 mm/mm) maximum; class B, .00050 inch/inch (0.00050 mm/mm) maximum.

After stress: Class A, .00075 inch/inch (0.00075 mm/mm) maximum; class B, .00050 inch/inch (0.00050 mm/mm) maximum.

Flexural strength (for base material thicknesses greater than or equal to .020 inch (0.51 mm)).

Crosswise: 50,000 pounds per square inch minimum.

Lengthwise: 60,000 pounds per square inch minimum.

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Flammability:

Duration of burning time, each flame application:

Base material thicknesses greater than or equal to .020 inch (0.51 mm).

Average: 25 seconds maximum

Single: 30 seconds maximum.

Base material thicknesses less than .020 inch (0.51 mm): 15 seconds maximum.

Burn length, average (for base material thicknesses less than .020 inch (0.51 mm)): 12 inches (305 mm) maximum.

Peel strength values for copper foil: See table I.

TABLE I. Peel strength values, minimum pounds per inch of width.

Test and copper foil weight	Base material thickness	
	Less than 0.020 inch	0.020 inch and over
After thermal stress:		
1/8-oz/ft^2 (designator E)	4.5	4.5
1/4-oz/ft^2 (designator Q)	5.0	5.0
3/8-oz/ft^2 (designator T)	5.5	5.5
1/2-oz/ft^2 (designator H)	4.5	6.0
3/4-oz/ft^2 (designator M)	5.0	7.0
1-oz/ft^2 (designator 1)	6.0	8.0
2-oz/ft^2 (designator 2)	8.0	11.0
3, 4, or 5 -oz/ft^2 (designator 3, 4, or 5)	9.0	12.0
At elevated temperature of 125°C:		
1/8-oz/ft^2 (designator E)	3.5	3.5
1/4-oz/ft^2 (designator Q)	4.0	4.0
3/8-oz/ft^2 (designator T)	5.0	5.0
1/2-oz/ft^2 (designator H)	4.0	4.0
3/4-oz/ft^2 (designator M)	4.5	4.5
1-oz/ft^2 (designator 1)	5.0	5.0
2-oz/ft^2 (designator 2)	6.0	6.0
3, 4, or 5 -oz/ft^2 (designator 3, 4, or 5)	7.0	7.0
After exposure to processing solutions:		
1/8-oz/ft^2 (designator E)	4.5	4.5
1/4-oz/ft^2 (designator Q)	4.5	4.5
3/8-oz/ft^2 (designator T)	5.0	5.0
1/2-oz/ft^2 (designator H)	3.0	4.5
3/4-oz/ft^2 (designator M)	4.0	5.0
1-oz/ft^2 (designator 1)	5.0	7.0
2-oz/ft^2 (designator 2)	6.0	9.0
3, 4, or 5 -oz/ft^2 (designator 3, 4, or 5)	7.0	12.0

Arc resistance: 60 seconds minimum.

Dielectric breakdown (for base material thicknesses greater than or equal to .020 inch (0.51 mm)): 40 kilovolts minimum

Electric strength, average, (for base material thicknesses less than .020 inch (0.51 mm)). 750 volts/mil minimum

Loss tangent at 1 MHz: 0.035 maximum.

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Permittivity at 1 MHz: 5.4 maximum.

Surface resistivity:

After moisture resistance conditioning: 10^4 megohm minimum.

After elevated temperature (125°C) conditioning: 10^3 megohm minimum.

Volume resistivity:

After moisture resistance conditioning: 10^6 megohm-cm minimum.

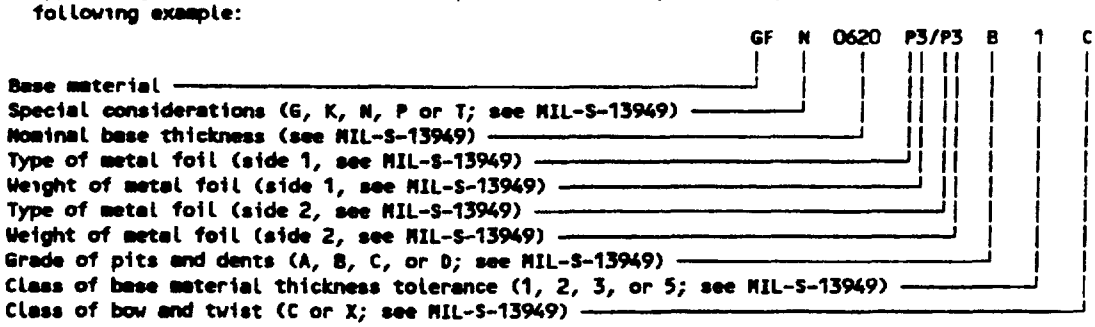
After elevated temperature (125°C) conditioning: 10^3 megohm-cm minimum.

Water absorption.

Not applicable for base material thickness less than .020 inch (0.51mm).
Base material thickness of .020 inch (0.51 mm) to .031 inch (0.79 mm): 0.80 percent.
Base material thickness of .032 inch (0.81 mm) to .062 inch (1.58 mm): 0.35 percent.
Base material thickness of .063 inch (1.60 mm) to .093 inch (2.36 mm): 0.25 percent.
Base material thickness of .094 inch (2.38 mm) to .125 inch (3.18 mm): 0.20 percent.
Base material thickness of .126 inch (3.20 mm) to .250 inch (6.35 mm): 0.13 percent.

Supersession information: Not applicable.

Type designation: Consists of "GF" prefix followed by coded alphas and numeric characters as shown in the following example:



Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES
NASA - NS

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5998-0024-01)

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